SIT233 – Cloud Computing
Task 4.1P: Creating an Amazon RDS database

Name: Nirosh Ravindran
CICRA Student ID: BSCP|CS|63|134

Task Questions

1. ## Question 1

   Answer: I would place it in the private subnet for enhanced security so only resources from the Virtual Private Cloud can access it and prevents any unauthorized access from the external network (internet). This feature is only beneficial if my RDS instance doesn't need much exposure from the external network to minimize any potential threats.

2. ## Question 2

   Answer: In the AWS Management Console, I will navigate to the Security Group associated with the RDS instance.

   In that we need to create a new inbound rule for the security group, like assigning port ranges that should be accessed, and specify the source IP address or range that our application server is hosted on.

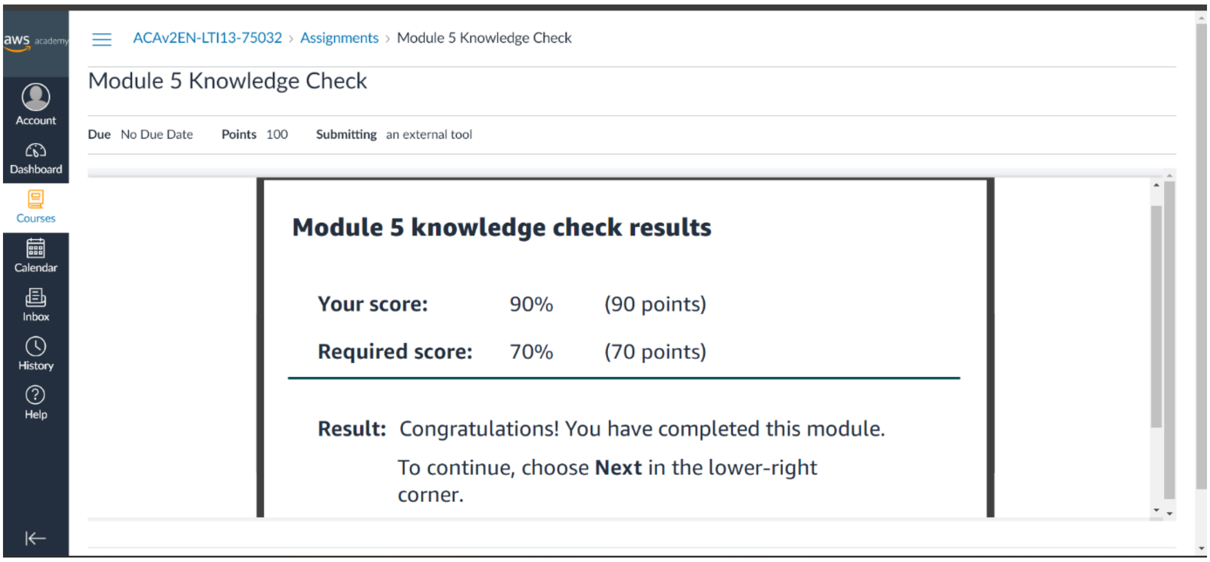
   By specifying these rules, we can make sure that the security group will only allow inbound traffic on the specified port only from the designated port only from the designated source, ensuring that only your application servers can connect to the RDS instance.

Quiz

1. Knowledge Check for Module 5

This knowledge check was comprised of the most important questions throughout the module 5. The questions asked was kind of a summary to the whole module 5. In this knowledge check I was able to see my understanding of this module, and the from the mistakes I was able to correct myself of the parts that I was not sure about.

This knowledge check was mainly about the relational and non-relational databases. There were questions about best suited database models according to the company needs, the types of databases, the benefits of each type, the use cases that each type of database is good at, and techniques used by the them and how they are used to migrate the on-premise data into cloud.



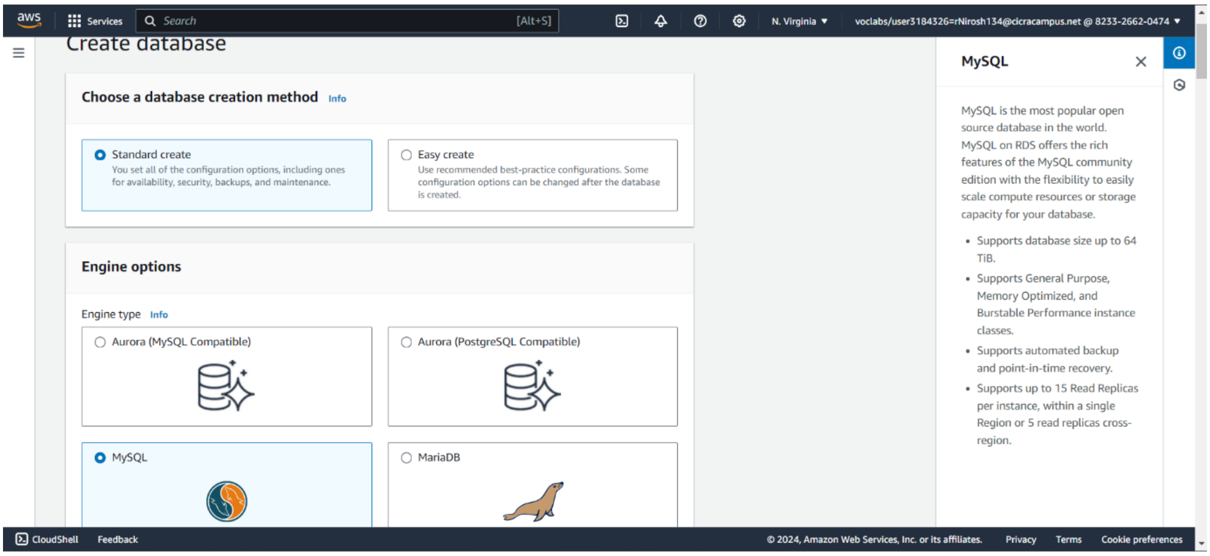
Module 5: Guided Lab – Creating an Amazon RDS Database

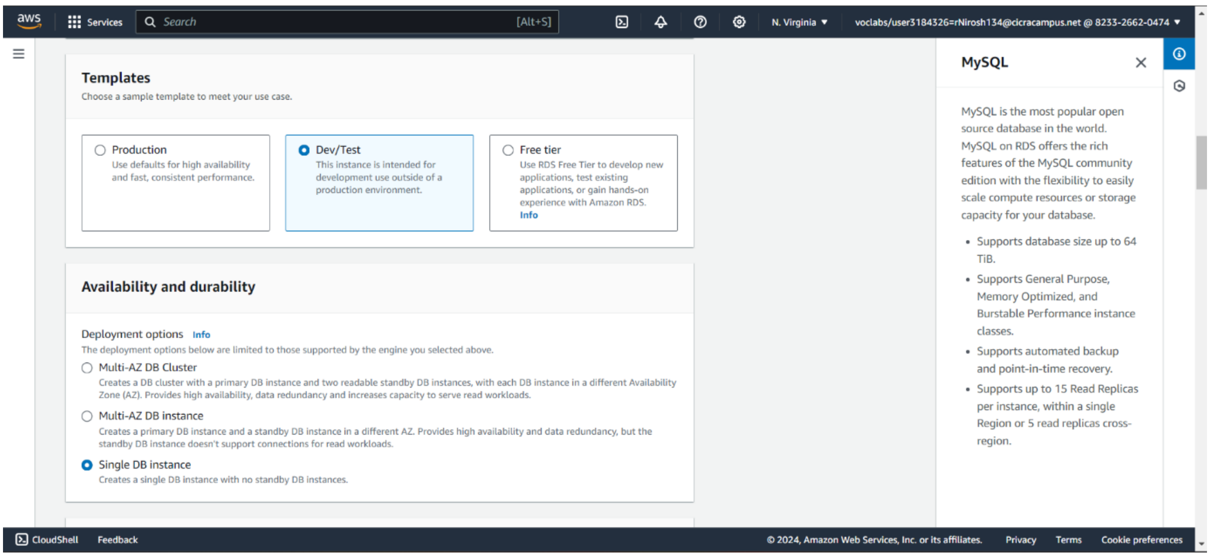
Task 1: Creating an Amazon RDS database

In the RDS console, I searched for "RDS" in the services search box and selected it. Then, clicked on "Create database."

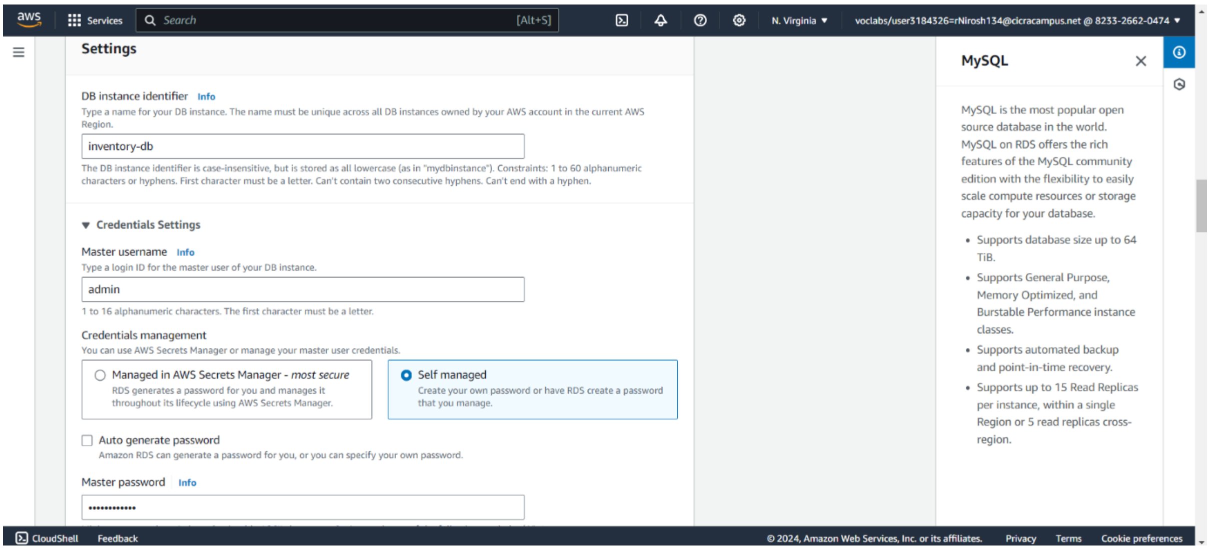
Under the "Engine options," I chose MySQL from the available options.

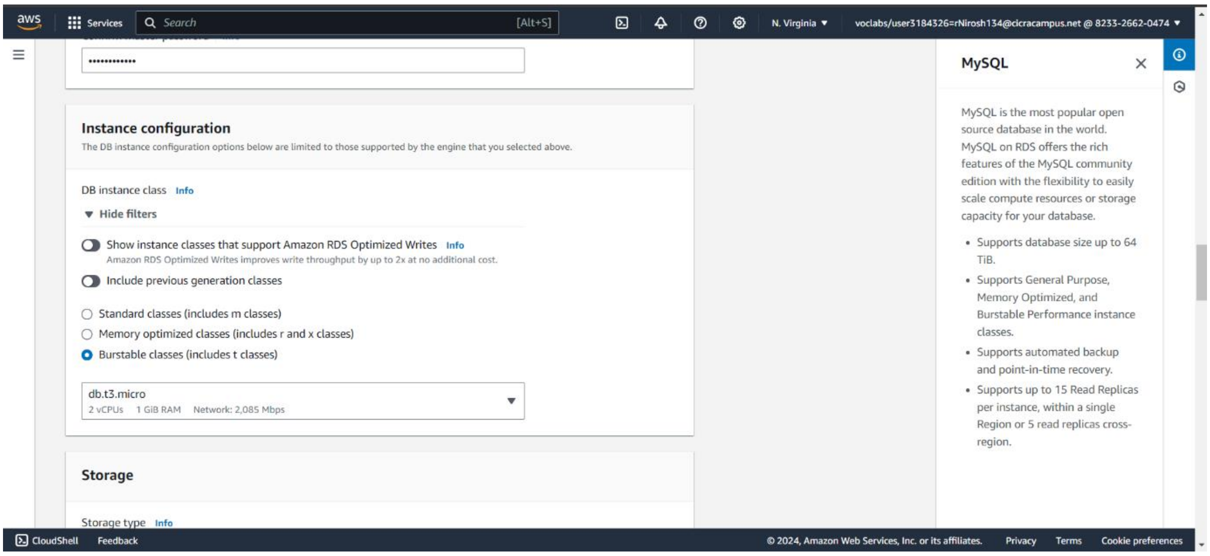
I Selected "Dev/Test" under the Templates section. Then Chose "Single DB instance" under the Availability and durability section.

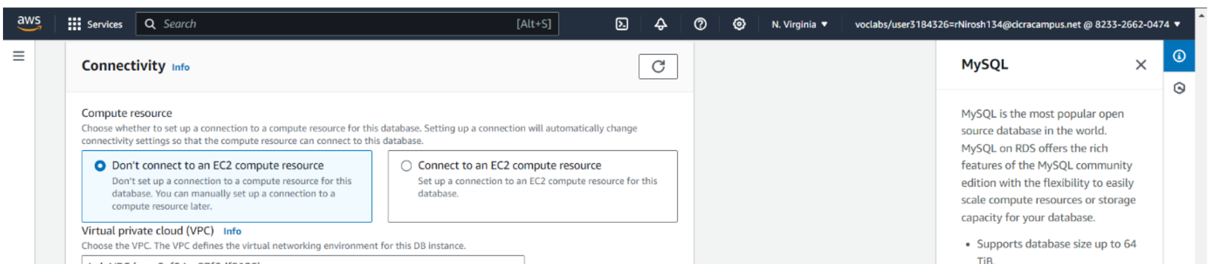


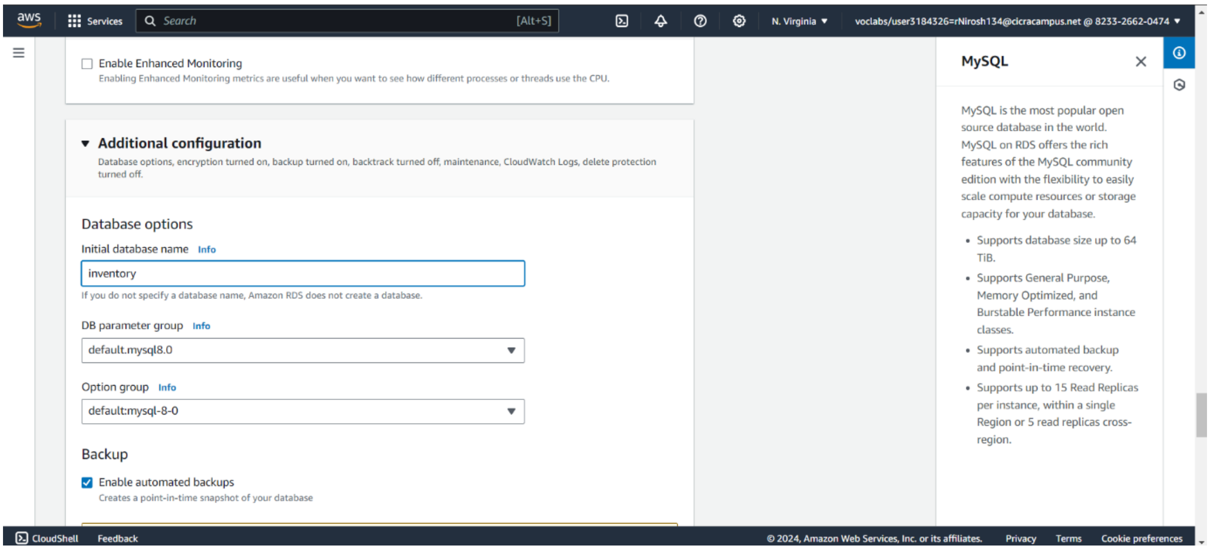


Then I configured some settings:
Set the DB instance identifier as "inventory-db."
Username as "admin."
Password as "lab-password" (confirm it).
Under DB instance class, select "Burstable classes (includes t classes)" and then choose "db.t3.micro."
For Connectivity, set the VPC to "Lab VPC" and select the existing VPC security group "DB-SG." Remove the default security group.
Set the initial database name as "inventory."
Clear (turn off) the "Enable Enhanced monitoring" option.

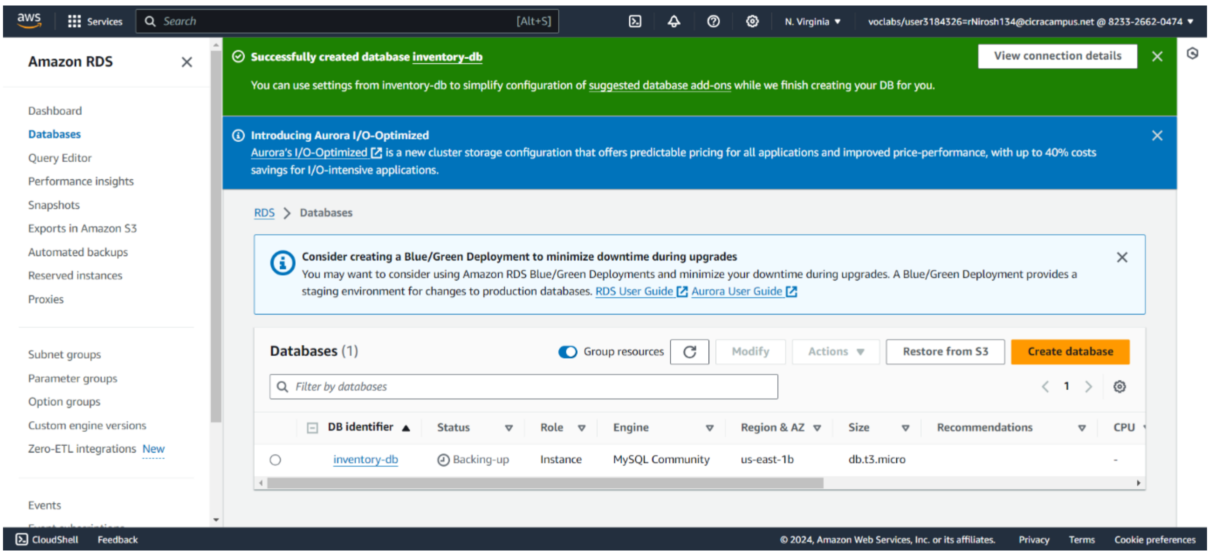






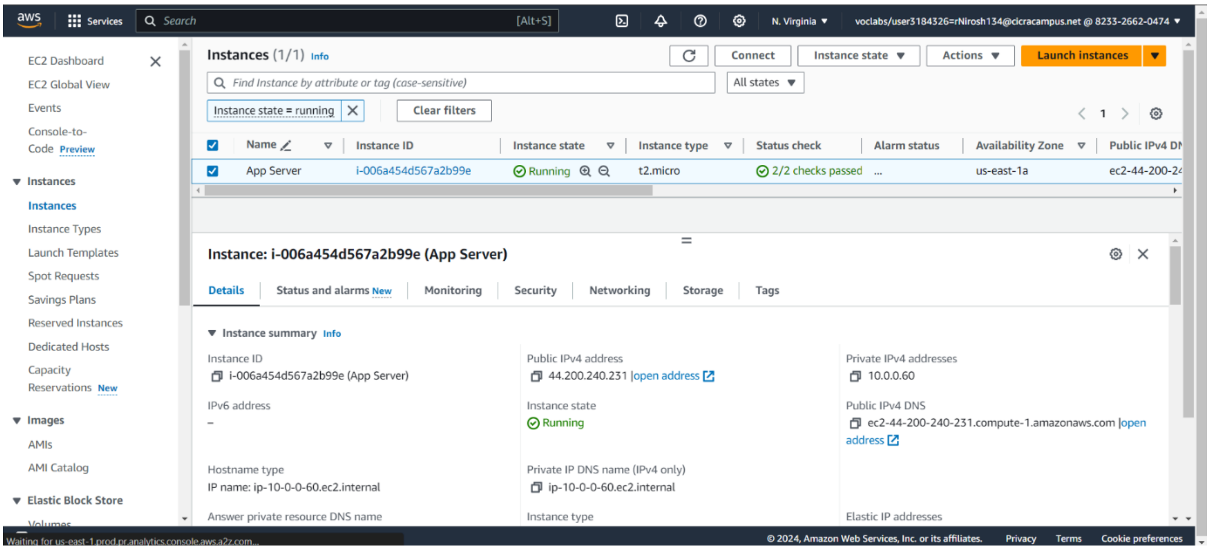


Then at last I clicked the create database and waited until the database was created. Finally, I received a message of successful database creation.

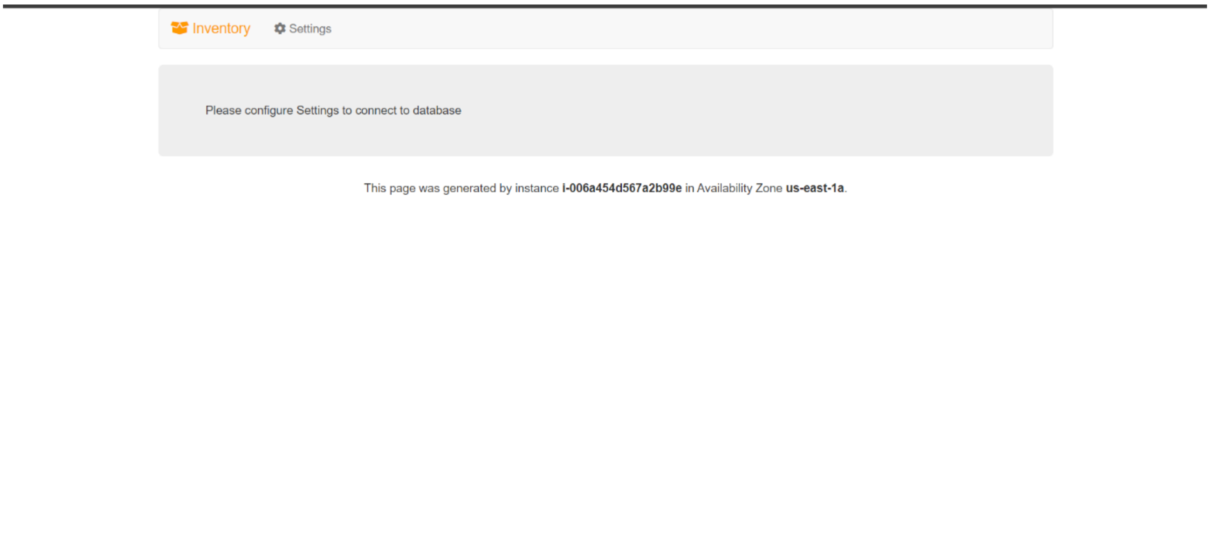


Task 2: Configuring web application communication with a database instance

I search for "EC2" in the services search box and choose EC2 to open the EC2 console. In the left navigation pane, I select "Instances." In the center pane, I locate a running instance named "App Server." I select the "App Server" instance and navigate to the "Details" tab. Hovering over the Public IPv4 address, I click on the copy icon to copy the address.



Opening a new web browser tab, I paste the copied IP address into the address bar and press ENTER. The web application loads, displaying limited information as it's not yet connected to the database.

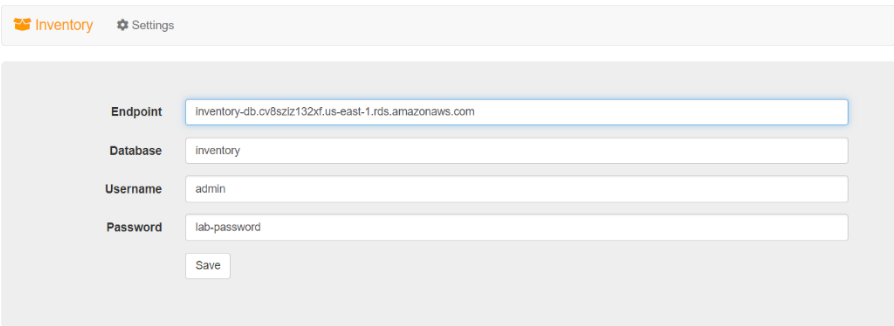


Name: Nirosh Ravindran
CICRA Student ID: BSCP|CS|63|134

Within the web application, I choose the "Settings" option to configure it to use the RDS DB instance created earlier.

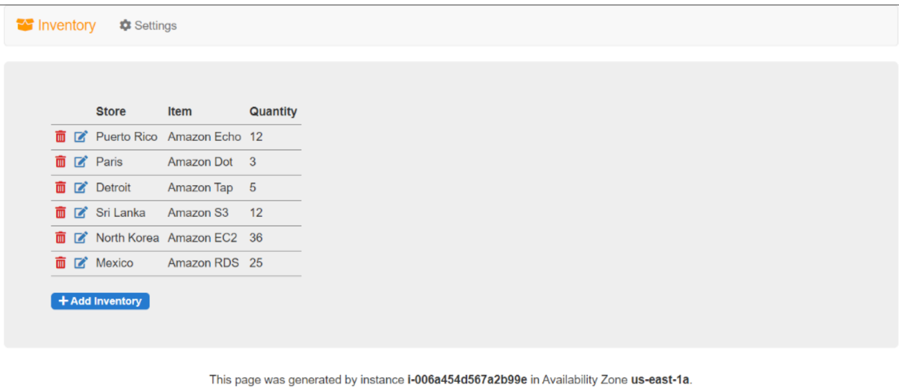
Returning to the AWS Management Console without closing the application tab, I search for and choose RDS to open the RDS console. In the left navigation pane, I select "Databases" and choose the "inventory-db." Scrolling to the "Connectivity & Security" section, I copy the Endpoint to my clipboard.

Back in the browser tab with the Inventory application, I enter the following values:
Endpoint: I paste the endpoint I copied earlier.
Database: I enter "inventory."
Username: I input "admin."
Password: I type in "lab-password."
After entering the details, I choose "Save."



With the provided database details, the application now connects to the database, loads initial data, and displays information.

I can now add, edit, and delete inventory information using the web application. Any changes made are stored in the Amazon RDS MySQL database created earlier, ensuring data integrity and availability even if there's a failure in the application server. I insert new records into the table to ensure there are at least 5 or more inventory records before completing the task.



Grade:

My Total score is 20/20